ASIF Siddique

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Date: 18/10/20/26

	outlook	Temperature	humidity	windy	Rain
	OL	T	H	W	R
1	Sunny	Hot	high	weak	No
2	Sunny	Mild	high	Strong	No
3	Cloudy	Hot	high	weak	Yes
4	Rainy	mild	high	weak	Yes
5	Rainy	Cool	normal	weak	yes
6	Rainy	Cool	Normal	Strong	No
7	cloudy	Mild	Normal	Strong	Yes
8	Sunny	Cool	Normal	Weak	No

Entropy of Outlook

Sunny = 3

Cloudy = 2 Total = 8

Rainy = 3

$H(S) = \sum -p_i \log_2(p_i)$

$-\frac{3}{8}\log_2\frac{3}{8} - \frac{2}{8}\log_2\frac{2}{8} - \frac{3}{8}\log_2\frac{3}{8}$

$-0.375\frac{\log(0.375)}{\log(2)} \quad -0.25\frac{\log(0.25)}{\log(2)} \quad -0.375\frac{\log(0.375)}{\log(2)}$

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Info gain of outlook

outlook	Total	Yes	No	Entropy
Sunny	3	0	3	0
Rainy	3	2	1	
Cloudy	2	2	0	0

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Date: / /

Entropy of Windy

weak = 5 Total = 8

strong = 3

$-\frac{5}{8}\log_2\frac{5}{8} \quad -\frac{3}{8}\log_2\frac{3}{8}$

$-(0.625)\frac{\log(0.625)}{\log(2)} - (0.375)\frac{\log(0.375)}{\log(2)}$

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Windy	Total	Yes	No
weak	5	3	2
Strong	3	1	2

Entropy of Rain

No = 4 Total = 8

Yes = 4

$-\frac{4}{8}\log_2\frac{4}{8} \quad -\frac{4}{8}\log_2\frac{4}{8}$

$-(0.5)\frac{\log(0.5)}{\log(2)} \quad -(0.5)\frac{\log(0.5)}{\log(2)}$

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DECENT

Date: / /

Entropy of Temperature

hot = 2

Mild = 3 Total = 8

Cool = 3

$$-\frac{2}{8}\log_2\frac{2}{8} - \frac{3}{8}\log_2\frac{3}{8} - \frac{3}{8}\log_2\frac{3}{8}$$

$$-\frac{(0.25)\log(0.25)}{\log(2)} - \frac{(0.375)\log(0.375)}{\log(2)} - \frac{(0.375)\log(0.375)}{\log(2)}$$

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Temp	Total	Yes	No
Hot	2	1	1
Mild	3	2	1
Cool	3	1	2

Entropy of Humidity

high = 4 Total = 8

Normal = 4

$$-\frac{4}{8}\log_2\frac{4}{8} - \frac{4}{8}\log_2\frac{4}{8}$$

$$-\frac{(0.5)\log(0.5)}{\log(2)} - \frac{(0.5)\log(0.5)}{\log(2)}$$

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Info Gain of Humidity

Humidity	Total	Yes	No	Entropy
High	4	2	2	
Normal	4	2	2	

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